

PROGRAMMED ARTICLE TRANSFER

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ABSTRACT

The present invention relates to the automatic operation of machinery, particularly to automatically operable materials handling apparatus, and to automatic control apparatus suitable for such machinery. A broad object of the present invention resides in the provision of article transfer apparatus having versatile program control means. The article transfer head 60 is moved by a mechanical power source through a sequence of strokes whose lengths and directions are determined by a program controller, here a program drum 62.

FIG. 2

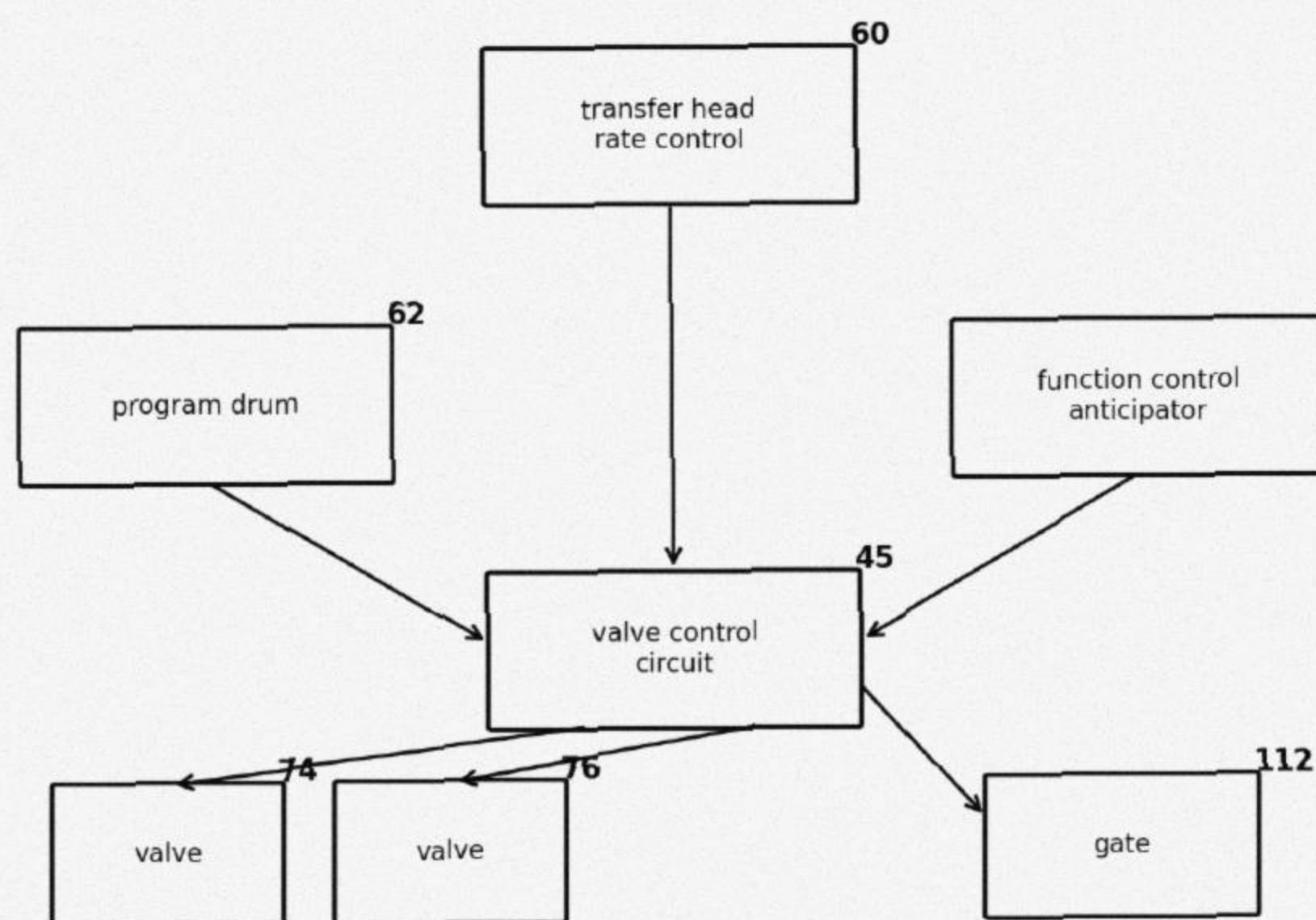


FIG. 2 — Transfer-head rate control, program drum 62, and valve control circuit 45.

BACKGROUND OF THE INVENTION

From the earliest years in the mechanical handling of articles there have been but two types of machine control: cam control and manual control. Manually controlled article handling equipment has the highly desirable property of being most flexible, but entails the high cost of operating manpower.

The present invention makes available for the first time a more or less general purpose machine that has universal application to a vast diversity of applications where cyclic control is desired. Universal automation, or Unimation, is a term that may well characterize the general object of the invention.

In applying one feature of this invention, the article transfer head 60 (whether it takes the form of jaws, a suction gripper, or other comparable article-handling tool) is moved by a mechanical power source through a sequence of strokes whose lengths and directions are determined by a program controller, i.e. program drum 62.

An especially desirable form of program controller combined with the transfer mechanism to be controlled represents a further feature of the invention. The transfer mechanism operates the transfer head and at the same time displaces a position detector; the position detector is compared through a feedback loop with the program controller until coincidence is reached.

A rate control 76 and a delayed-gate valve control circuit 45 cooperate with a function control anticipator 64 so that fast traverse of the transfer head is reduced in advance of the required stopping point, avoiding overtravel. Valves 74 and 76 admit hydraulic fluid to the actuating cylinder under control of gate 112.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a plan view, partly diagrammatic, showing one form of programmed article transfer apparatus embodying the invention.

FIG. 2 is a block diagram of the transfer-head rate control, program drum 62, function control anticipator 64, and valve control circuit 45 with valves 74 and 76 and gate 112.

FIG. 3 is a diagrammatic view of the recording channel and program drum of the control apparatus.

CLAIMS

What is claimed is:

1. In a programmed article transfer apparatus, the combination comprising: an article transfer head 60 movable through a sequence of strokes; a program drum 62 storing a recorded sequence of position codes; a function control anticipator 64 coupled to the program drum for reducing a traverse rate of the transfer head in advance of a required stopping point; and a valve control circuit 45 responsive to the function control anticipator for actuating a pair of valves 74, 76 to admit fluid to a cylinder driving the transfer head.
2. The apparatus of claim 1, further comprising a gate 112 interposed between the valve control circuit 45 and the valves 74, 76, the gate being responsive to a delayed-gate signal generated by the function control anticipator 64.
3. The apparatus of claim 1, wherein the program drum 62 comprises a recording channel on which is recorded a sequence of coded positions corresponding to successive

strokes of the transfer head 60.

4. The apparatus of claim 1, wherein the transfer head 60 comprises a gripper for grasping an article and the sequence of strokes comprises lifting, swinging and twisting motions each controlled by a separate track of the program drum 62.